Environmental Science & Technology

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Electric Power Reliability, Energy Burdens, and Climate Change Beliefs in the United States

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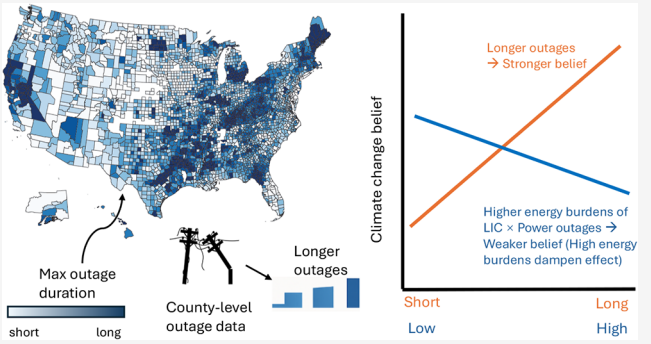
Cite This: https://doi.org/10.1021/acs.est.5c09116

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ABSTRACT: The electric power system is essential to social and economic well-being but remains vulnerable to climate-related hazards that exacerbate power outages. This study asks: Are the duration of power outages and the extent of energy burdens (EBs) associated with public beliefs about climate change across the United States? While previous research identified media exposure, demographics, and geography as influences on climate change beliefs, the role of power outages and EBs has not been examined. Using county-level geospatial regression analysis, we find a significant positive association between power outages and climate change beliefs across the continental United States, with longer outages associated with greater beliefs. Each one-unit increase in maximum outage duration corresponds to a 5.16% increase in beliefs, accounting for EB, media exposure, political affiliation, and education. However, the interaction between power outages and EBs reveals a negative relationship with climate change beliefs in three out of the four regions, namely, the South, West, and Midwest, indicating higher energy costs can dampen the belief-enhancing effect of outages. These findings highlight that power outages and EBs are important correlates of climate change beliefs and have region-specific implications for climate action policy and public communication amid mounting climate-related disruptions.



KEYWORDS: *climate change beliefs, climate change, power outages, energy burdens, geospatial regression, extreme weather*

1. INTRODUCTION

The electric power system, commonly referred to as the grid, represents a form of critical infrastructure underpinning social and economic wellbeing.[1−3] However, electricity reliability is increasingly susceptible to disruptions from climate change such as more severe weather and other hazards, impinging on energy security and exacerbating community vulnerability.[4−7] These weather-related power outages already cause up to $70 billion in annual inflation-adjusted losses across the United States (U.S.),[8] costs borne by the public in the form of lost wages, spoiled inventory, delayed production, inconvenience, and direct physical damage to assets. These electricity blackouts also have potent negative health impacts, including carbon monoxide poisoning and temperature-related illnesses, and can be fatal for individuals who depend on electricity-powered medical devices at home or who face high energy burdens (EBs).[9−11]

This study investigates whether the severity of power outages and the extent of household EBs are associated with public beliefs about climate change across the U.S. Prior work has largely examined technical dimensions of grid failures, such as outage causation and infrastructure resilience[5,6] or the socioeconomic drivers of EB,[12] with limited attention to whether public climate change beliefs fit into this picture. Despite extensive research[13] documenting associations between beliefs and social,[14−16] psychological,[17−19] political,[20−23] and demographic factors (education,[23] age,[24] gender,[25] etc.), the interplay between energy factors, such as power outages and high energy costs, land climate change beliefs remains underexplored. Meanwhile, climate change has increasingly caused more frequent and intense power outages worldwide,[26−28] disrupting communities and economies, yet little is known about whether such experiences shape public beliefs.

Understanding this relationship has significant social and environmental implications. Identifying the roles of outages and EBs, exacerbated by climate-driven events like hurricanes, heatwaves, and wildfires, could inform more targeted risk communication and policy strategies to enhance climate awareness and resilience.[29,30] Conversely, ignoring spatial dynamics and environmental triggers risks underestimating the societal impacts of climate change and misallocating

Received: July 5, 2025
Revised: November 22, 2025
Accepted: November 24, 2025

ACS Publications

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resources for mitigation and adaptation.[31,32] To address this gap, we examine the association among climate change beliefs, power outages, and EBs across U.S. regions. Our analysis employs geospatial regression methods to capture spatial dependencies overlooked in prior ordinary least squares (OLS)-based work.[33] We further explore how demographic and socioeconomic factors, including political affiliation, education, age groups, and minority status, mediate this relationship, revealing the geographic and social dimensions of belief formation.

2. BACKGROUND AND LITERATURE REVIEW

2.1. Climate Change Beliefs. Climate change beliefs are multidimensional and defined as a range of public attitudes and risk perceptions regarding global warming—beliefs about whether it is happening, the extent to which humans cause it, and the personal and societal risks it poses.[34] In the U.S., they are often measured using the data from the Yale Climate Opinion Maps (YCOM), which provide county-level estimates of beliefs, risk perceptions, and policy preferences, derived from large-scale national surveys combined with demographic and geographic modeling.[34,35] Specifically, YCOM's six indicators capture public climate change beliefs: (1) **Happening**—the percentage of adults who believe global warming is occurring; (2) **Human**—the percentage who attribute global warming primarily to human activities; (3) **Consensus**—the percentage who agree that most scientists believe global warming is occurring; (4) **Affecting weather**—the percentage who perceive global warming as affecting U.S. weather patterns; (5) **Experience**—the percentage who report personally experiencing the effects of global warming; and (6) **Priority**—the percentage who consider a presidential candidate's stance on global warming important to their vote. Together, these indicators capture both cognitive (occurrence, attribution, scientific agreement) and experiential (perceived weather effects, personal experience) and political (vote priority) aspects of public perceptions. For methodological consistency, we treat climate change and global warming as interchangeable terms, reflecting YCOM's original operationalization of the construct as global warming.

A growing body of research examines the determinants of climate change beliefs, highlighting the roles of social, psychological, political, and demographic factors. For instance, conservative ideology is consistently associated with lower belief in climate change, whereas higher education and minoritized status often correlate with stronger belief.[36] Education also interacts with ideology,[37] shaping how information is interpreted, and minority communities frequently express greater concern than white individuals, in part due to disproportionate environmental risks and lived exposure.[38]

Leiserowitz's foundational surveys[39] show that risk perceptions and cultural values such as worldviews strongly shape U.S. climate change concern; McCright and Dunlap[36] document the ideological polarization of climate change beliefs, showing widening partisan divides over time. Howe et al.[14] extended this work by applying multilevel regression with poststratification to generate county-level belief estimates, forming the basis of the YCOM conducted by the Yale Program on Climate Change Communication (YPCCC). International comparisons, such as a study conducted by Lee et al.,[40] reveal substantial cross-national variation, with higher concern in Europe and Latin America than in parts of Asia and the U.S., underscoring the importance of context in shaping perceptions. Within Latin America, a recent study reports similar patterns in which conservative ideology and lower education predict reduced belief, while minority status predicts stronger belief.[33]

Despite these advances, much of the work relies on cross-sectional national surveys and linear regression methods (e.g., OLS) that fail to account for geographic clustering, nonlinearities, and spatial dependencies inherent in belief patterns.[33,41] Aggregated survey approaches can obscure the local heterogeneity. Moreover, while social and political correlates are well established, the potential association with infrastructure stressors and energy-related hardship remains largely unexamined. To our knowledge, no systematic framework explicitly links infrastructure disruptions, such as power outages or household EBs, to climate change beliefs, leaving a critical gap at the intersection of infrastructure, lived experience, and public opinion.

2.2. Power Outages and Hazard Experience. Climate change is expected to increase both the frequency and severity of power outages. Grid vulnerabilities arise from a range of hazards: extreme cold temperatures elevate generator forced outage rates; stronger winds during storms or hurricanes damage overhead lines via debris or collapsing pylons and towers; flooding impairs substation equipment; and wildfires destroy transmission corridors.[42−45] Power outage events can be characterized using two standard metrics: outage duration (the length of time customers are without service) and peak-outage prevalence (the share of customers without power at the event peak), as illustrated in Supplementary Figure S2. These measures are widely used in engineering and resilience studies, but their potential to explain social and perceptual outcomes remains underexplored.

A growing literature shows that direct exposure to climate hazards shapes social beliefs, policy preferences, and support for low-carbon technologies.[9,46,47] Personal experience with extreme weather and natural disasters tends to heighten concern about climate action or increase support for progressive energy and climate policies.[48−51] This pattern is evident even among minority groups and Indigenous communities. However, the literature has focused more broadly on hazard exposure than specifically on infrastructure disruptions.

Studies centered explicitly on power outages are comparatively limited. For example,[28] researchers examined California's 2019 public safety power shut-offs and found effects on behavioral intentions, climate attitudes, utility perceptions (i.e., attitudes toward utilities), and government trust (i.e., politician approval ratings). Yet, most existing studies rely on aggregate national survey data[41] or single-state cases, overlooking spatially explicit factors such as local outage experience that may shape belief formation. At the national scale, the U.S. Department of Energy (DOE)'s The Environment for Analysis of Geo-Located Energy Information (EAGLE-I) data set provides county-level outage data from 2014 to 2023 and is widely used in engineering analyses of grid reliability.[52,53] To date, applications have largely focused on physical infrastructure resilience with far less attention to the social or perceptual dimensions of disruption. This gap motivates our approach: integrating infrastructure disruptions, not merely hazard exposure, in the study of climate change beliefs and doing so with methods that explicitly account for spatial dependence and clustering in U.S. counties.

https://doi.org/10.1021/acs.est.5c09116
Environ. Sci. Technol. XXXX, XXX, XXX−XXX

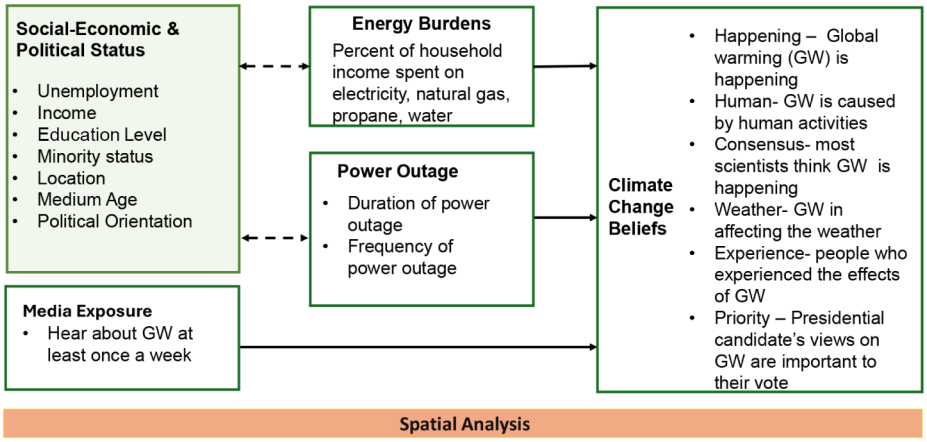


Figure 1. The conceptual framework of key variables.

2.3. Energy Burden. Energy burden (EB), defined as the percentage of household income spent on energy costs, represents a more precise measure of energy-related hardship and insecurity than income alone.[54,55] In some countries, EBs are so serious, and pronounced, that they become affiliated with the term "energy poverty" or "fuel poverty," referring to those who spend more than 10−15% of their monthly income on heat or electricity services.[56,57] This focus on a financial definition of burden corresponds with the fact that income reflects resources available, whereas EB measures the claim utility costs make on those resources, more closely reflecting households' day-to-day constraints. Yet most studies use income as a proxy for economic constraint, overlooking EB as a direct measure of household energy hardship. This gap matters: high burdens force trade-offs among essentials (e.g., housing, healthcare, utilities), which can diminish attention to long-term environmental risks and reduce environmental engagement when immediate survival needs take precedence.[55,58,59]

The problem of high EBs is exacerbated by extreme weather. Households have to operate their air conditioners for longer periods of time during heat waves and face greater heating needs during cold spells. Additionally, severe storms have the potential to disrupt the power supply, resulting in price increases or outages. For already overburdened households, these shocks intensify difficult choices between meeting other basic needs and keeping the lights on. According to recent research, temperature extremes and climate change cause temporary increases in household energy expenditures.[60] Low-income renters and communities of color are particularly affected because they are less likely to have effective cooling systems, adequate insulation, or backup power and hence have reduced productivity.[61]

EB is central to the broader concept of energy insecurity, which encompasses not only affordability but also reliability, quality of service, and the ability to meet basic household needs.[62] In major U.S. cities, low-income, Black, Latino, and renter households bear disproportionately high burdens.[63] These burdens cluster spatially across neighborhoods,[64] highlighting the geographic variation of these inequities. Moreover, county-level evidence shows that EBs cluster in areas with social disadvantage, compounding racial and health disparities.[65] Recent work further conceptualizes energy insecurity as a multidimensional condition, including economic, physical, and behavioral components.[66]

At the county scale, the DOE's Low-Income Energy Affordability Data (LEAD) Tool provides estimates of EBs by income group, fuel type, and housing characteristics.[67] This data set supports assessments of the distribution of energy costs across U.S. households and informs policy discussions on energy justice.

Despite this policy salience, the EB literature has not directly linked this form of economic stress to climate change belief. Given evidence that financial strain can suppress environmental engagement when immediate needs overshadow broader concerns,[59] clarifying this relationship is essential to understanding how material hardship shapes public beliefs.

2.4. Gaps in Knowledge. Taken together, prior research supports three core insights. First, demographic and social forces, including partisanship, ideology, education, race/ethnicity, and community context, are well-established determinants of climate change beliefs. Second, EB is a salient marker of household hardship with plausible implications for how people appraise climate and health risks over time. Third, climate-driven hazards and grid disruptions (e.g., heat waves, floods, and blackouts) can shape attitudes and policy preferences in specific settings.

However, to the authors' knowledge, no prior study has systematically integrated these strands to examine whether power outages and EBs are jointly associated with climate change beliefs. Existing work is often limited to single states, small case studies, or national surveys that fail to account for spatial dependencies in belief formation. This gap motivates our study, which applies geospatial regression methods to assess the relationship among outages, EBs, and climate beliefs while explicitly modeling spatial clustering across U.S. counties. This approach advances the literature by unifying social-demographic determinants with infrastructure stressors, improving inference on place-based dynamics, and informing policy designs that couple energy affordability, reliability, and climate communication. Figure 1 illustrates the conceptual framework and summarizes the key variables in our analysis.

3. MATERIALS AND METHODS

This study used several nationally representative data sources. We used the county as our unit of analysis for the following four main reasons. First, outage data from the EAGLE-I data set are reported only at the county level, which sets the finest defensible spatial resolution for our study. Second, the YCOM provides climate belief estimates at the county scale—the

https://doi.org/10.1021/acs.est.5c09116
Environ. Sci. Technol. XXXX, XXX, XXX−XXX

smallest consistently released geography.[14] Third, county-level socio-demographic, EBs, and infrastructure indicators are systematically reported and widely used in resilience and policy planning.[68] Fourth, counties are policy-relevant jurisdictions that span and differentiate across urban, suburban, and rural contexts.[69] We acknowledge that aggregation to the county scale can mask neighborhood-level disparities (i.e., potential Modifiable Areal Unit Problem and ecological-inference concerns); however, this scale enables consistent regional comparisons while summarizing intracounty variation in energy and climate vulnerability.[64,70]

3.1. Climate Change Beliefs in the U.S. For the climate change beliefs measure, we utilize the YCOM data developed by the YPCCC, which provides estimates of U.S. climate change beliefs, risk perceptions, and policy preferences at the state and local levels.[34] The estimates were derived from a statistical model using multilevel regression with poststratification (MRP) on a large national survey data set ($n > 31{,}000$), along with demographic and geographic population characteristics. The raw data were collected through large-scale surveys, such as the "Climate Change in the American Mind" series. These surveys were conducted with nationally representative samples of U.S. adults, using both online and telephone-based methods. The program collaborates with experienced survey partners to ensure robust sampling and rigorous data collection techniques. These surveys have become a leading resource for understanding public attitudes toward climate change in the U.S.

In our analyses, we used the county-level climate change beliefs data from 2018 to 2023. It provides an estimated percentage of U.S. adults' beliefs on climate change in terms of six dimensions: Happening, Human, Consensus, Affecting Weather, Experience, and Priority. The estimations of climate change beliefs were based on public responses to relevant survey questions. For mapping purposes, response categories from multiple questions were consolidated into a single variable. For example, for the question measuring respondents' belief in whether global warming is affecting weather in the U.S., responses of "Strongly agree" and "Somewhat agree" were combined into a single "Agree" category, while "Somewhat disagree" and "Strongly disagree" were grouped into a single "Disagree" category. These consolidated categories were then translated into numeric values to create county-level proportions for each belief indicator, ensuring comparability across geographies while preserving the ordinal structure of the responses.

To account for potential noise and inconsistencies in survey responses, arising from respondent bias or measurement error, we constructed a composite climate change belief variable. By aggregation of multiple related metrics, this approach enhances reliability and mitigates the influence of outliers and response inconsistencies. The six belief indicators capture distinct yet interconnected dimensions of climate change perception, including its occurrence, anthropogenic causes, and perceived priority. Rather than treating these aspects in isolation, a single aggregated variable provides a more comprehensive representation of climate change beliefs. The constructed belief metric, B_y^i, is defined as

$$B_y^i = \lambda_1 \cdot \text{Happening}_y^i + \lambda_2 \cdot \text{Human}_y^i + \lambda_3 \cdot \text{Consensus}_y^i + \lambda_4 \cdot \text{Affect weather}_y^i + \lambda_5 \cdot \text{Experience}_y^i + \lambda_6 \cdot \text{Priority}_y^i \quad (1)$$

where i and y denote the county and year indices, respectively. Factor analysis was employed to determine data-driven weights (λ_1 to λ_6) based on the variance explained, rather than assuming equal contributions from each indicator. This method is widely utilized in social science research to derive factor weights that reflect the relative importance of indicators in capturing the underlying construct.[71] Internal consistency was assessed using Cronbach's alpha ($\alpha = 0.99$), confirming the robustness of the composite measure.[72] A Cronbach's alpha below 0.70 would have warranted reconsideration of the variable aggregation. For the climate change beliefs data in 2023, the calculated weights were $\lambda_1 = 0.1735$, $\lambda_2 = 0.1701$, $\lambda_3 = 0.1636$, $\lambda_4 = 0.1696$, $\lambda_5 = 0.1600$, and $\lambda_6 = 0.1631$. This composite index differs from a simple numeric recoding of survey responses as the factor weights (λ_1–λ_6) reflect empirically derived contributions of each belief dimension to the underlying construct.

3.2. Power Outage Events in the U.S. For the power outage events analysis, we utilized the EAGLE-I data set,[52] which provides county-level power outage information at 15 min intervals over recent years. Our study period spans 2018–2023. The data set covers a wide range of geographic regions across the U.S. identified by their Federal Information Processing Standards (FIPS) codes. Key attributes include time-stamps marking the start of outages (*run_start_time*) and the number of affected customers (*sum*), recorded at 15 min intervals. No data are recorded if there is no outage. The EAGLE-I data set does not track individual outage; instead, it only contains the total number of affected customers in a county at a given time. Although several studies have used the EAGLE-I data set to study power system resilience,[53] there lacks a systematical method on extracting critical information such as detailed outage events, their corresponding durations, and metrics on outage intensity (e.g., the maximum number of customers impacted per event) for each event. To fill this gap, we developed a comprehensive EAGLE-I data processing procedure to systematically extract and analyze these outage characteristics.

Supplementary Figure S3 presents the data processing procedure for the EAGLE-I data set which is designed to transform raw outage records into insightful analyses that enhance our understanding of outage events. The process begins with the integration of geographical and historical coverage data. By importing county-level shapefiles (population data included) and merging them with coverage history from CSV files, we can calculate the number of customers in each county as

$$\text{Customers}_y^{CL,i} = \frac{\text{Population}_y^{CL,i}}{\text{Population}_y^{SL,i}} \times \text{Customers}_y^{SL,i} \quad (2)$$

where superscripts *CL* and *SL* represent the county-level and state-level values. eq 2 allocated the state-level customer counts to individual counties based on population distribution for the subsequent outage analysis. Once county-level customer data are obtained, the procedure proceeds to clean and preprocess the raw outage data. The raw outage records, spanning from 2018 to 2023, were imported from CSV files and sorted by state and county (or FIPS) to maintain county-level consistency. Missing values were addressed by introducing zeroes at 15 min intervals where no outages are recorded, ensuring a comprehensive data set free of gaps. This step is critical for maintaining the integrity of the time-series analyses

https://doi.org/10.1021/acs.est.5c09116
Environ. Sci. Technol. XXXX, XXX, XXX−XXX

that follow. The data were then enriched by merging them with preprocessed geographical data, which includes total customer counts for each county. This enables the calculation of outage intensities relative to population sizes.

A critical aspect of the processing involves identifying and grouping distinct outage events. In this study, we define a county-level power outage event as occurring when the maximum outage percentage exceeds a specified threshold of the total customers in the county. Counts of customers without power were obtained from the EAGLE-I dataset , and the percentage of county customers without power (in 15 min resolution) was estimated based on state-level total service customer data and county-level population information from the 2023 Census Bureau. This estimation is expressed by the following equation:

$$P^{i}_{\mathrm{out},t} = \frac{C^{i}_{\mathrm{out},t}}{\mathrm{Customers}_{y}^{CL,i}} \tag{3}$$

where $P^{i}_{\mathrm{out},t}$ is the percentage of customers without power at time t for county i. $C^{i}_{\mathrm{out},t}$ is the number of customers without power at time t for county i. We applied a 1.66% threshold (i.e., relative intensity metric), corresponding to the 99th percentile of customers without power per 15 min interval at the county level during the 2023 study period. Thus, for a county with 10,000 customers, our power outage definition is met when at least 166 customers lose power. This percentile-based threshold is designed to identify the most severe 1% of events, which are typically characterized by prolonged duration, a large fraction of affected customers, and a high degree of public salience. Prior studies have used the 90th percentile to determine outage events.[26] However, our analysis of the EAGLE-I data set reveals that a significant portion of outages are transient events, lasting only minutes and typically arising from routine grid operations rather than extreme weather. This work focuses specifically on weather-induced outages. Consequently, the 99th percentile was selected to filter out these frequent but less impactful incidents and isolate the rare, high-impact outages caused by severe weather that are disproportionately salient in shaping community awareness.[73] This approach is consistent with percentile-based indices in climate science (e.g., R95/R99 for precipitation and temperature extremes) and with established reliability practices, such as the IEEE Std. 1366 for designating "major event" days from the statistical tail of outage distributions.[74] The county-level power outage characteristics of the customers in the U.S. in the past 5 years are presented in Supplementary Table S3.

Outage events were assessed against predefined relative intensity metrics to isolate the most impactful ones. This ensures that only significant outages exceeding the established thresholds are retained for further analysis. Next, the system assigned sequential outage numbers within each county, making it easier to track and reference distinct outage events. The processed outage data were then analyzed for duration, calculating the length of each outage event. By aggregating outage durations and applying minimum duration thresholds, the workflow identified sustained power disruptions that require deeper investigation. These duration metrics, combined with intensity measures, provide a comprehensive view of the outage landscape and highlight areas with recurrent or prolonged power issues. Finally, the workflow concluded with the visualization and storage of the processed data.

3.3. Energy Burdens in the U.S. For the most recent year with available EB data, this study used the 2022 version of the Low-Income Energy Affordability Data (LEAD) Tool.[67] It provides nationally consistent estimates of household EB at state, county, and census tract levels, based on factors such as income, energy expenditures, fuel type, housing type, and geography. The estimates are generated by integrating household demographic and income data from the American Community Survey (ACS) with housing and building stock information from the Residential Energy Consumption Survey (RECS), and then applying modeled energy expenditures from DOE's Residential Energy Consumption Model.

This approach enables estimation of household energy costs as a share of income across different income brackets within each geography, yielding a distribution of EBs at the county level. While the LEAD tool offers standardized, policy-relevant, and geographically detailed data, the estimates are modeled outputs rather than direct county-level survey responses and, therefore, may contain uncertainty. Nevertheless, its integration of ACS and RECS data makes it one of the most comprehensive and widely used resources for evaluating energy affordability across the U.S. In our regression analyses, we used the EBs of low-income communities (LIC), defined as households earning less than 80% of Area Median Income, as one of the independent variables.

3.4. Linear Regression Analysis. To evaluate the relationship between changes in power outage durations and shifts in climate change beliefs within the same geographic areas over time, we adopted a first-differences methodology inspired by reference.[75] This approach compares each location to itself across different time periods, offering two key advantages: 1) it accounts for unobserved, time-invariant confounders specific to each FIPS code, and 2) it minimizes dependence on the absolute values of maximum power outage durations by focusing on the relative ranking of changes over time.

For each year t, we categorized FIPS codes into M = 40 quantiles based on their change in maximum power outage duration from 2018 to year t, represented as $T^{\Delta}_{p,t}$. Concurrently, we calculated the change in climate change beliefs for each FIPS code over the same period, denoted as B^{Δ}_{t}. Using eq 4, we examined the relationship between these two variables to determine whether counties experiencing greater increases in power outage durations also saw more pronounced shifts in climate change beliefs. The analysis controlled for baseline covariates from 2018 ($X_{q,2018}$) and their changes since then ($X^{\Delta}_{q,t}$), with all covariates standardized by mean-centering. The equation used for estimation is

$$B^{\Delta}_{t} = \sum_{p=1}^{M} \alpha_{p} T^{\Delta}_{p,t} + \sum_{q=1}^{N} \beta_{q} X_{q,2018} + \sum_{q=1}^{N} \beta^{\Delta}_{q} X^{\Delta}_{q,t} + \epsilon \tag{4}$$

In this linear regression study, we included multiple independent variables, including percentage of unemployment, educational attainment (percentage of the population with no high school diploma or a bachelor's degree or higher), age groups (18−34, 35−54, and 55+ years), gender composition (percentage of females), and minority status.

3.5. Geospatial Regression Analysis. To account for spatial dependencies and heteroskedasticity in the data, we employed the Generalized Method of Moments (GMM)-based spatial autoregressive combined model with hetero-

https://doi.org/10.1021/acs.est.5c09116
Environ. Sci. Technol. XXXX, XXX, XXX−XXX

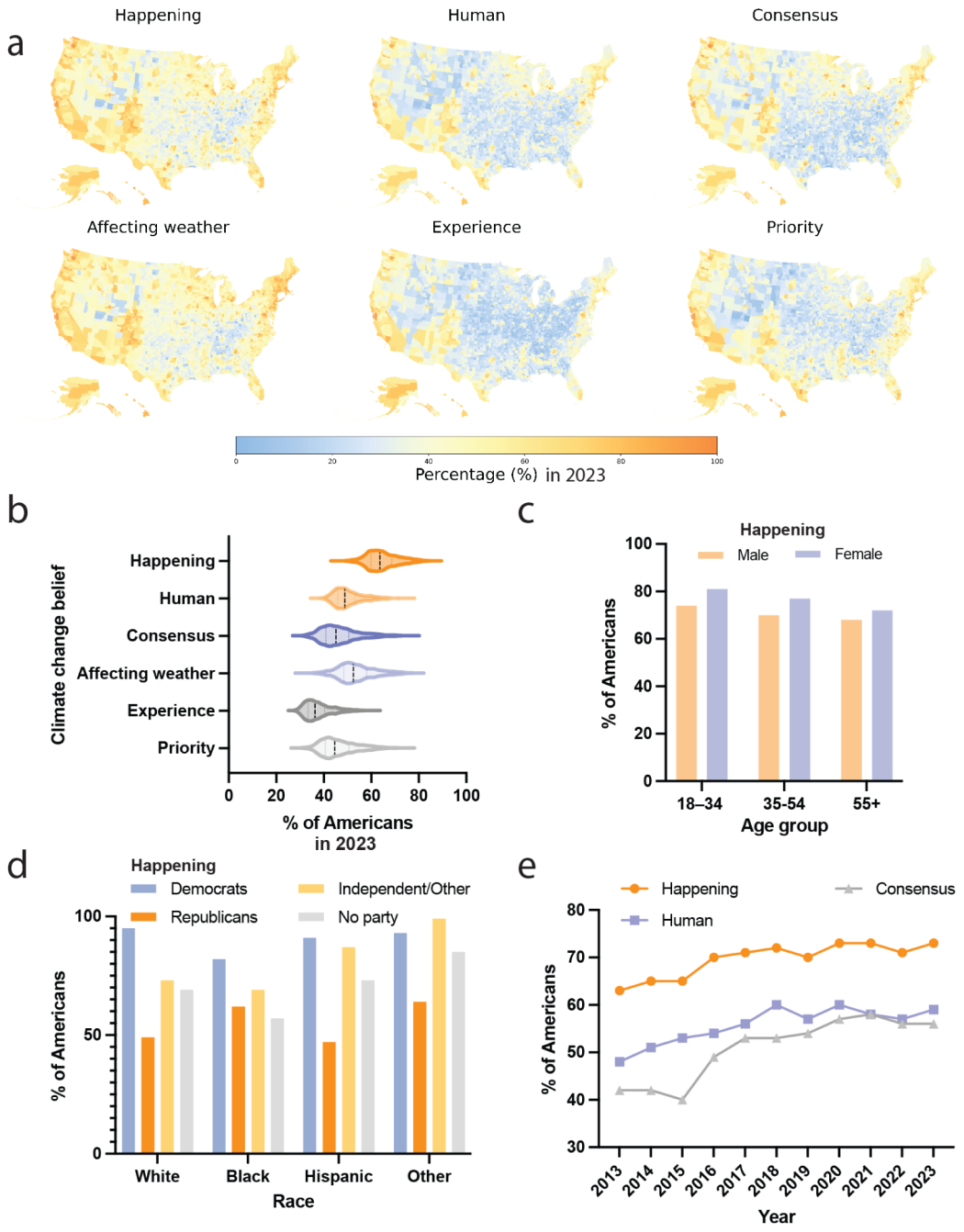


Figure 2. Summary of climate change beliefs across the United States. (**a**) County-level estimated percentage of adults in 2023 who believe climate change which measured by six indicators. (**b**) Distribution of county-level climate change beliefs in 2023. (**c**) Climate change beliefs (Happening) by age and gender. (**d**) Climate change beliefs (Happening) by race and political party. (**e**) National average climate change beliefs over time.

skedasticity (GM_Combo_Het) from the spreg package (version v1.8.4.dev20).[76] This model extends traditional regression by incorporating both spatial lag (ρ) and spatial error (λ) components, capturing spatial spillover effects in both the dependent variable and the error structure. The model is specified as

$$B = \rho \cdot W \cdot B + X \cdot \beta + u,\ u = \lambda \cdot W \cdot u + \epsilon \quad (5)$$

where B is the dependent variable (climate change beliefs), X represents the matrix of independent variables, W is the spatial weights matrix, and u denotes spatially autocorrelated errors. The error term ϵ is assumed to be heteroskedastic. The GM_Combo_Het estimator corrects for spatial dependence by using an instrumental variable approach, ensuring consistent and efficient parameter estimation under heteroskedasticity. This model is particularly suitable for analyzing geographically structured survey data as it accounts for spatial interactions that may influence climate change beliefs across neighboring regions.

3.5.1. Dependent Variables. The dependent variables of the geospatial regression model include the constructed climate change beliefs metric in eq 1 and the six indicators (i.e., Happening, Human, Consensus, Affecting weather, Experience, and Priority). The estimated values of these seven climate change belief metrics for each county in 2023 were obtained from the YCOM data set.

https://doi.org/10.1021/acs.est.5c09116
Environ. Sci. Technol. XXXX, XXX, XXX−XXX

3.5.2. Independent Variables. The independent variables in the geospatial regression model capture socioeconomic, demographic, and ideological factors that may influence climate change beliefs. These include percentage of unemployment, educational attainment (percentage of the population with no high school diploma or a bachelor's degree or higher), age groups (18−34, 35−54, and 55+ years), gender composition (percentage of females), minority status, political affiliation (percentage of votes for the Republican candidate in the most recent presidential election), and media exposure (percentage of adults who hear about global warming in the media at least once a week). Additionally, we incorporated measures of EBs of low-income communities and maximum power outage durations. To examine potential interactive effects, we included an interaction term between EBs and maximum power outage durations. Spatially lagged versions of key predictors account for potential regional spillover effects. To ensure comparability and mitigate scaling issues, all independent variables were standardized.

Most variables (e.g., power outage, media weekly, no high school, bachelor's degree, age groups, minority status, Republican affiliation) show statistically significant coefficients across the regression models (Supplementary Tables S4−S10), indicating robust relationships with dependent variables.

4. RESULTS

Based on our extensive methods, this section presents our core findings, organized around the five themes of trends in beliefs, power outage events, temporal associations, geospatial analysis, and regional variation.

Initially, we used linear regression (e.g., ordinary least-squares) and found that climate change beliefs are not linearly related to power outages, highlighting the limitations of nonspatial methods due to significant spatial autocorrelation. Our geospatial analyses further reveal that longer power outage durations are significantly associated with higher climate change beliefs. The interaction effect of power outage duration and EB showed a significant relationship with climate change beliefs in some regions. Leveraging Moran's I test and geospatial regression models, we identify spatial clusters of heightened belief and prolonged outage duration in regions like California and the Northeast, confirming strong spatial dependence overlooked by traditional OLS models. These findings underscore that power outages and EBs are significant factors associated with climate change beliefs, offering novel insights into the interplay between environmental disruptions and public perception in a global warming world.

4.1. Trends in Climate Change Beliefs. Using the 2023 YCOM data,[34] we assessed six key indicators of public climate change beliefs across U.S. counties: Happening, Human, Consensus, Affecting weather, Experience, and Priority. These indicators constitute "Climate Change Beliefs", one of five variable categories in the YCOM data set.

Our analysis reveals substantial geographic and demographic variability in these beliefs, reflecting the complex and heterogeneous nature of public opinion on climate change. Geographically, we observed pronounced disparities in climate change beliefs across U.S. counties (Figure 2a). In 2023, an average of 64.6% of Americans believed that global warming is happening, with county-level agreement ranging from 45.4% to 87.2% (Figure 2b). However, belief levels vary considerably, depending on the specific aspect of climate change being considered. For example, agreement that global warming is affecting U.S. weather patterns is lower and more polarized, ranging from 30.6% to 79.4% at the county level. Similarly, belief in the scientific consensus on climate change shows an even wider range, from 29.6% to 77.5%, reflecting varying levels of trust in scientific institutions across counties. This wide range highlights the diversity of public opinion on climate change, even within the same country.

Coastal counties, particularly in the West and Northeast, exhibited higher levels of belief across all six dimensions compared to inland regions. Conversely, counties in the Midwest and South showed lower levels of belief, especially regarding personal experiences of global warming's effects. The spatial variability in climate change beliefs underscores the importance of the regional context in shaping public perceptions. Factors such as exposure to climate-related impacts, race, gender, education, and age may all contribute to these differences (see Figures 2c,d and Supplementary S1). Furthermore, the distinct ranges and patterns of belief across the six Climate Change Belief indicators, such as Happening, Human, and Consensus, highlight the complexity of public opinion on climate change. For instance, while most people agree that global warming is happening, fewer are convinced of its human origins or the scientific consensus, pointing to potential gaps in the public understanding of climate science.

To capture temporal variation, we assessed climate change beliefs from 2013 to 2023 and found a steady increase, reflecting growing public awareness and concern about global warming (Figure 2e). During this period, the percentage of Americans who agree that climate change is happening rose from 63% to 73%. Despite this progress, significant disparities persist across demographic groups. Younger adults (18−34 years) and women are more likely to acknowledge climate change and its impacts compared to older adults and men, respectively (Figure 2c). Similarly, income, racial, and political ideology plays a significant role, with Democrats and white populations expressing greater concern about climate change (Figures 2d and Supplementary S1).

4.2. Power Outage Events across U.S. Regions. For the most recent year with available power outage data, 2023, our study encompassed 2,925 counties, revealing varying levels of power outages across different regions, as illustrated in Figure 3. The median county-level annual power outage count ranged between 20 and 26 events per year, while the median county-level maximum outage duration ranged between 1.25 and 14.25 h per year. At the state level, Hawaii (mean = 52), Louisiana (mean = 48), California (mean = 48), Maine (mean = 47), and Texas (mean = 43) had the highest average outage event counts, while Maine (mean = 85 h), Hawaii (mean = 78 h), California (mean = 62 h), Kentucky (mean = 59 h), and Mississippi (mean = 47 h) experienced the longest average maximum outage duration (Supplementary Tables S1−S2).

Spatially, power outage events concentrated in the Southeast, Northeast, Appalachia, and parts of California. Notably, regions with frequent power outages often experienced prolonged outage durations, as well. For example, in Maine, the mean annual maximum outage duration reached 85.1 h in 2023, while the mean annual outage count was 47.1—placing it among the top 5 states for both metrics. In other words, power outages in these regions were more severe than in other parts of the country, both in terms of outage frequency and maximum outage duration. From 2018 to 2023, the statistical characteristics of power outage events in the U.S. did not show

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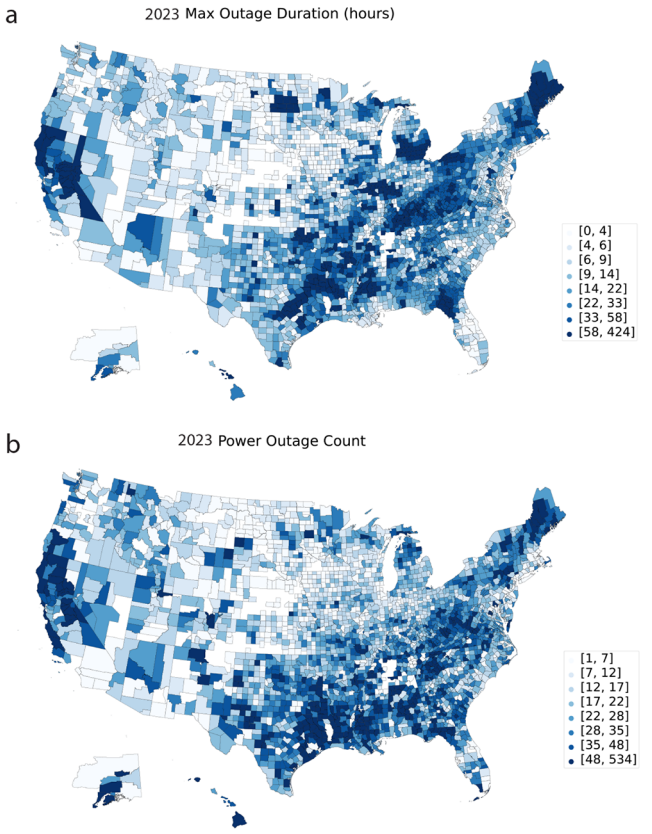


Figure 3. County-level power outage events reported in 2,925 counties across the United States. (**a**) Geographic distribution of county-level maximum power outage durations in 2023. (**b**) Geographic distribution of county-level power outage event counts in 2023.

significant changes at the national level, as shown in Supplementary Table S3.

4.3. Temporal Associations between Outages and Climate Change Beliefs. Our analysis identified a notable overlap between regions experiencing severe power outages and areas with a high prevalence of climate change beliefs. To evaluate temporal association, we used multiple linear regression models (Section 3.4) to analyze whether changes in maximum power outage durations from 2018 to 2023 were associated with shifts in climate change beliefs. Figure 4 illustrates this relationship by categorizing the changes of the county-level annual maximum outage duration into 40 quantiles. Our results indicate a general decline in climate change beliefs (Human) from 2020 to 2023 compared to 2018 levels, a trend that aligns with the curve depicted in Figure 2c. However, the data patterns illustrated in Figure 4 reveal an important nuance: the magnitude of this decline in climate change beliefs (Human) was less pronounced in counties that experienced larger increases in annual maximum power outage durations.

This linear relationship (blue lines in Figure 4) was statistically significant only for the year 2020 ($p < 0.05$), when counties in the highest quantile saw no net change (0.0%) in belief, compared to a −0.3% change in counties with the lowest quantile. This suggests that, during 2020, communities experiencing more severe power outages were more likely to report heightened concern about climate change. The linear association was not statistically significant in 2021, 2022, or 2023 (Figures 4 and Supplementary S4−S6). However, the variability in the relationship across years suggests it may not be strictly linear. This raises the possibility that nonlinear regression methods could provide a more nuanced understanding of these dynamics.

4.4. Geospatial Correlations among Climate Change Beliefs, Power Outages, and Energy Burdens. To further explore the possible nonlinear association between climate change beliefs and power outages, we employed geospatial regression analysis, which accounts for spatial dependencies and heterogeneity in the data. We constructed a composite climate change belief metric via factor analysis (Section 3) of all six indicators. Moran's I statistic (0.62) suggests strong spatial clustering in belief levels between the attribute and its spatial lag (Supplementary Figure S7).

The geospatial correlation analysis between the constructed climate change belief metric and 13 independent variables provides key insights into the interplay among power outage durations, EBs, socio-demographic factors, political influences, and public perceptions of climate change. Regression results (Figure 5) show that belief levels are significantly associated with several independent variables. Across the 2,925 counties analyzed in the continental U.S., a one-unit increase in the annual maximum power outage duration is associated, on average, with a 5.16% point increase in the proportion of Americans who believe in climate change ($p < 0.01$). Media exposure and minority populations share were also positively correlated, while Republican affiliation was negatively correlated. EBs of LICs, defined as the proportion of household income spent on energy costs, were not significantly associated with climate change beliefs. However, an interaction term between power outages and EBs of LIC showed a significant negative association with the beliefs, suggesting the impact of direct energy-related hardships in shaping public awareness.

Disaggregating by individual belief indicator (Figure 6), we found that "Human" and "Consensus" dimensions had the strongest positive correlations with power outages, media exposure, and minority populations. A one-unit increase in the annual maximum power outage duration was associated with an 8.40-point increase in the proportion of those who believe climate change is caused by human activities ($p < 0.01$) and a 6.27% point increase in the proportion of Americans who agree that most scientists believe global warming is occurring ($p < 0.01$). The "Happening" and "Experience" dimensions also exhibited significant positive associations with these factors (especially power outages). Moreover, we found that the "Human", "Consensus", "Happening", and "Experience" dimensions exhibited negative correlations with the interaction term between power outages and EBs of LIC. However, the "Affecting weather" and "Priority" dimensions yield mixed results, with factors such as power outages, unemployment, and education levels (Bachelor's degree) showing non-significant associations with the beliefs, suggesting that the dimensions may be influenced by more complex, multifaceted influences.

4.5. Regional Variation in Climate Change Beliefs and Energy Factors. To examine the regional difference, we estimated separate geospatial regression models for the four U.S. Census regions (see Figure 7a). The regions follow the U.S. Centers for Disease Control and Prevention (CDC) classification, which groups states into South, West, Midwest, and Northeast. The results presented in Figure 7b reveal distinct regional variations in the association between climate

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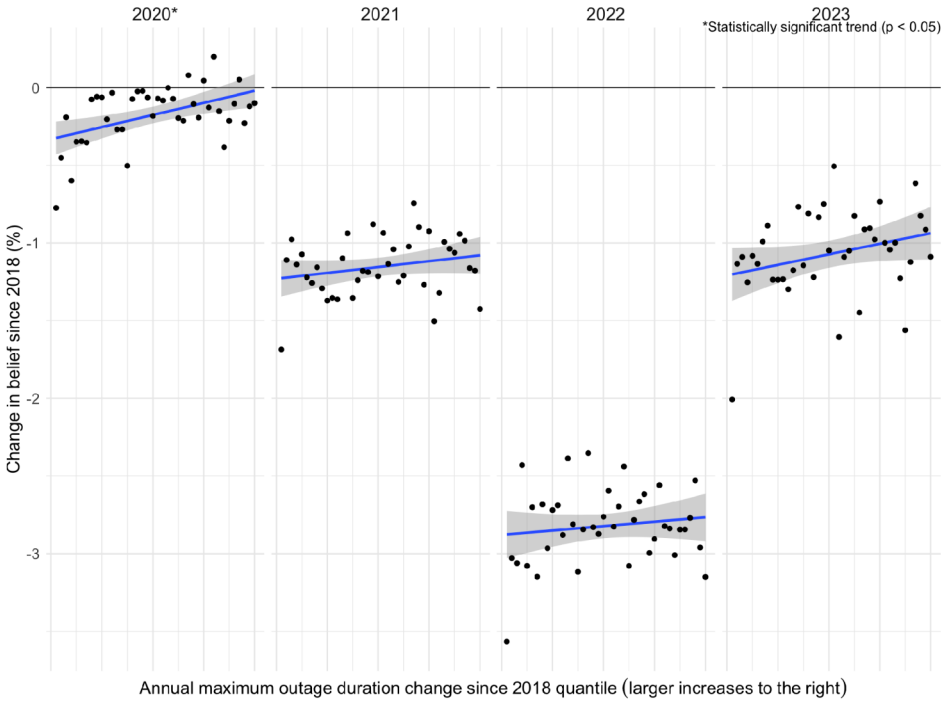


Figure 4. County-level changes in belief (Human), categorized by quantiles (α_p) of maximum outage duration changes, as estimated by our linear regression model (see eq 4 in Methods). Positive values on the vertical axis denote increases in climate change beliefs. Panel labels (2020, 2021, 2022, and 2023) denote years of comparison to 2018. Higher quantile numbers represent counties with greater increases in power outage durations, while the *y*-axis reflects the percentage change in climate change beliefs (Human) relative to 2018 levels.

change beliefs and the 13 selected independent variables. In the South ($p < 0.05$), West ($p < 0.05$), and Midwest ($p < 0.01$), longer maximum outage durations were significantly associated with stronger climate change beliefs (+5.12, +9.83, and +13.32% points, respectively). In contrast, in the Northeast, higher EBs of LIC were associated with lower belief levels, indicating potential trade-offs between economic stress and environmental concern. This suggests that higher energy costs may shift low-income household priorities away from climate concerns toward immediate financial pressures. This aligns with prior findings indicating that economic constraints often limit public engagement with long-term environmental issues.[58,59]

The interaction between power outages and EB showed a significant negative correlation with beliefs in three of the four regions, including the South, West, and Midwest regions. Conversely, the maximum duration of power outages exhibited a significant positive correlation with beliefs in the same three regions. Across all four regions, Republican affiliation maintained a consistent negative association with climate change beliefs. These regional disparities underscore the complex and context-dependent nature of climate change perception, emphasizing the need for localized energy resilience strategies and targeted communication efforts to address region-specific concerns.

5. DISCUSSION

To the best of our knowledge, this study presents the first nationwide geospatial analysis of how power outages and EBs interconnect with public climate change beliefs across the U.S. Our findings reveal a nonlinear and regionally heterogeneous relationship. Prolonged power outages are generally associated with stronger climate change beliefs, especially in the South, West, and Midwest, but this pattern is attenuated or reversed in areas with high EBs. These results challenge a common assumption in climate risk perception: direct experience with climate-linked stressors invariably increases the level of concern. Instead, our results suggest a contextual interpretation of the infrastructure disruptions. In some regions, outages are viewed as evidence of a changing climate. However, in high-EB communities, the same stressors appear to be interpreted through the lens of economic precarity and systemic inequity. Residents may attribute grid failures and high bill costs to immediate infrastructural or policy failures,[59] eroding trust and the cognitive bandwidth needed to engage broader environmental narratives. This points to a contextual theory of climate belief formation, where the meaning of an event such as a blackout is shaped by local socioeconomic conditions. Consequently, our findings imply that policy interventions cannot be one-size-fits-all. Effectively building climate engagement may first require addressing the foundational issues of energy justice and grid equity. Policies aimed at improving reliability and reducing cost burdens are not just infrastructural goals but are potentially crucial for fostering public capacity to engage with climate change. In other words, progressive energy policy implementation and local low-carbon transitions will only move at the speed of trust.[77]

While a majority of Americans now recognize the reality of climate change, the persistence of demographic and geographic divides underscores the need for targeted communication strategies and policy interventions to address these disparities. Understanding these trends is critical for fostering broader public engagement and support for climate action. For instance, as shown in Figure 4, communities that experienced more severe power outages showed relatively smaller declines in climate change beliefs, suggesting that localized climate-related stressors can heighten public awareness—even during

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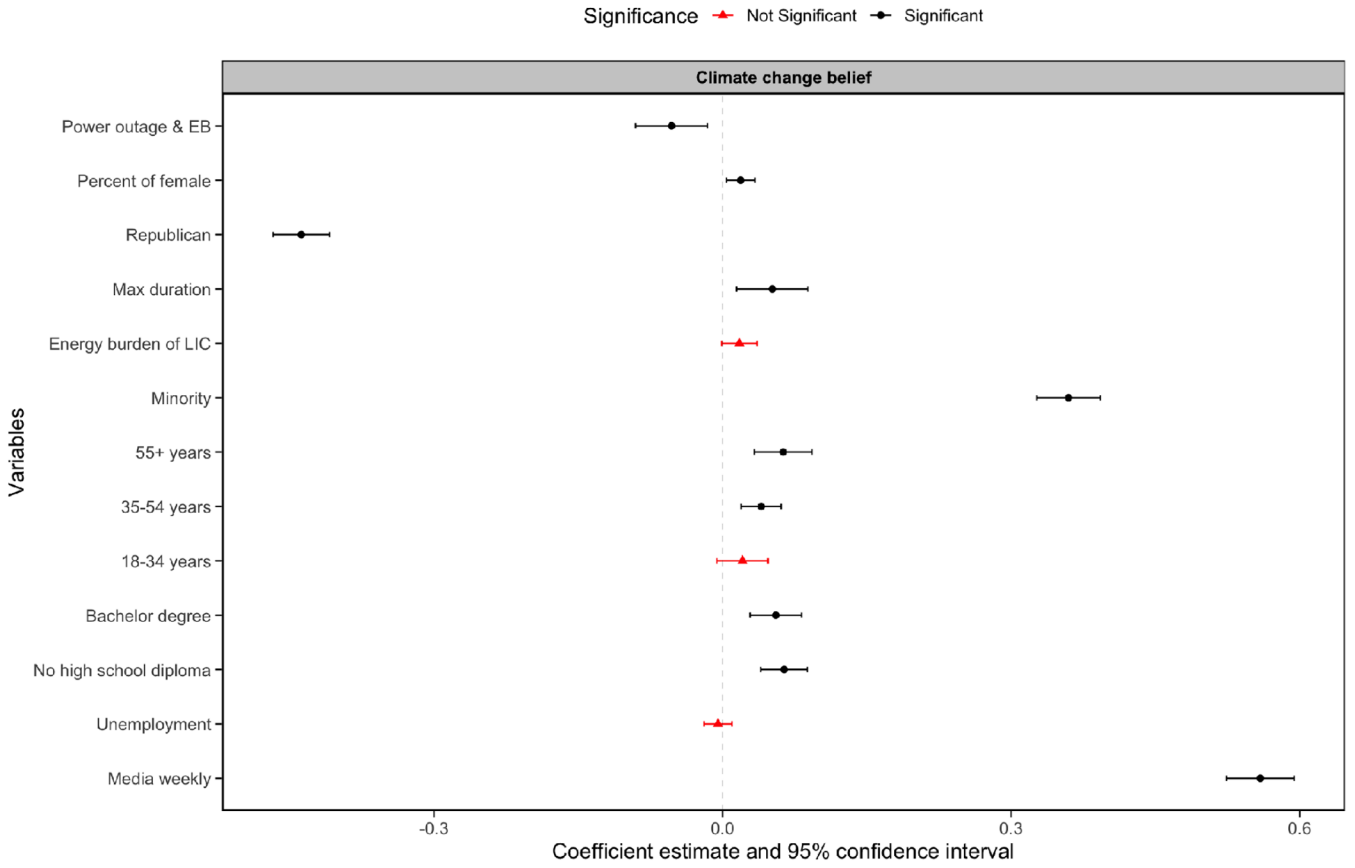


Figure 5. Associations between American beliefs in climate change and 13 independent variables. The climate change belief metric is an aggregated metric of the six belief indicators. Points are adjusted coefficient estimates from the geospatial regression model. Error bars correspond to 95% Confidence Intervals (CIs). Dotted lines denote the null of no association. Red bars indicate that the corresponding independent variable is not significantly correlated with the climate change belief metric, while black bars indicate a significant correlation. EB is energy burden. Max duration is maximum power outage durations. Power outage & EB is the interaction term between power outages and EB of low-income communities.

periods when national belief levels declined, such as from 2020 to 2023. However, this effect appeared strongest in 2020, indicating that shocks may generate short-term surges in salience that wane over time. This temporal fragility raises a critical question: do outages and disruptions function as catalysts for long-lasting shifts in climate belief, or do they serve as episodic reminders whose influence fades once normalcy is restored? Future research should explicitly test this by examining how long such belief shifts persist and whether they are reinforced or weakened by subsequent policy action, community resilience measures, or partisan narratives. If the salience of shocks is fleeting, then durable engagement will likely require institutional reinforcement that links disruptions directly to climate narratives.

Our geospatial regression model corroborates prior work, showing that political affiliation, education, and minority status strongly shape climate change beliefs. Conservative ideology and lower education attainment are associated with reduced belief, while minority status is positively associated, consistent with earlier findings.[33,37,38] The negative relationship between Republican affiliation and climate change beliefs across all four census regions further underscores the need for bipartisan approaches in climate communication and policymaking. The juxtaposition here is especially important: while infrastructural stressors such as outages and EBs do relate to climate belief patterns, an entrenched political identity often exerts a stronger and more consistent effect. This suggests that purely technical improvements, such as making the grid more reliable, will not automatically translate into greater public engagement unless the narratives surrounding these improvements bridge partisan divides. A thought-provoking implication is that energy infrastructure resilience is necessary but not sufficient; political communication strategies must operate in tandem to ensure that infrastructure experiences reinforce, rather than compete with, broader climate messaging.

In contrast to prior studies that use income as a proxy for economic constraint, we emphasize EBs as a more precise measure of energy-related hardship. High EB soften force trade-offs among essential services like housing, healthcare, and utilities, which in turn shape perceptions of long-term environmental risks.[58,59] This measure captures the materiality of energy experiences in a way that income cannot: it reflects the lived trade-offs households make each month. Our findings suggest that when these constraints are acute, attention to climate change may recede, not because residents deny environmental risks, but because immediate survival takes precedence. This finding challenges the assumption that awareness always increases with hardship and instead suggests that hardship can polarize engagement, pushing some groups toward climate change while pushing others toward disengagement.

Importantly, our study also suggests that no single factor shapes climate change beliefs in isolation. Significant correlations with power outages, climate-related media

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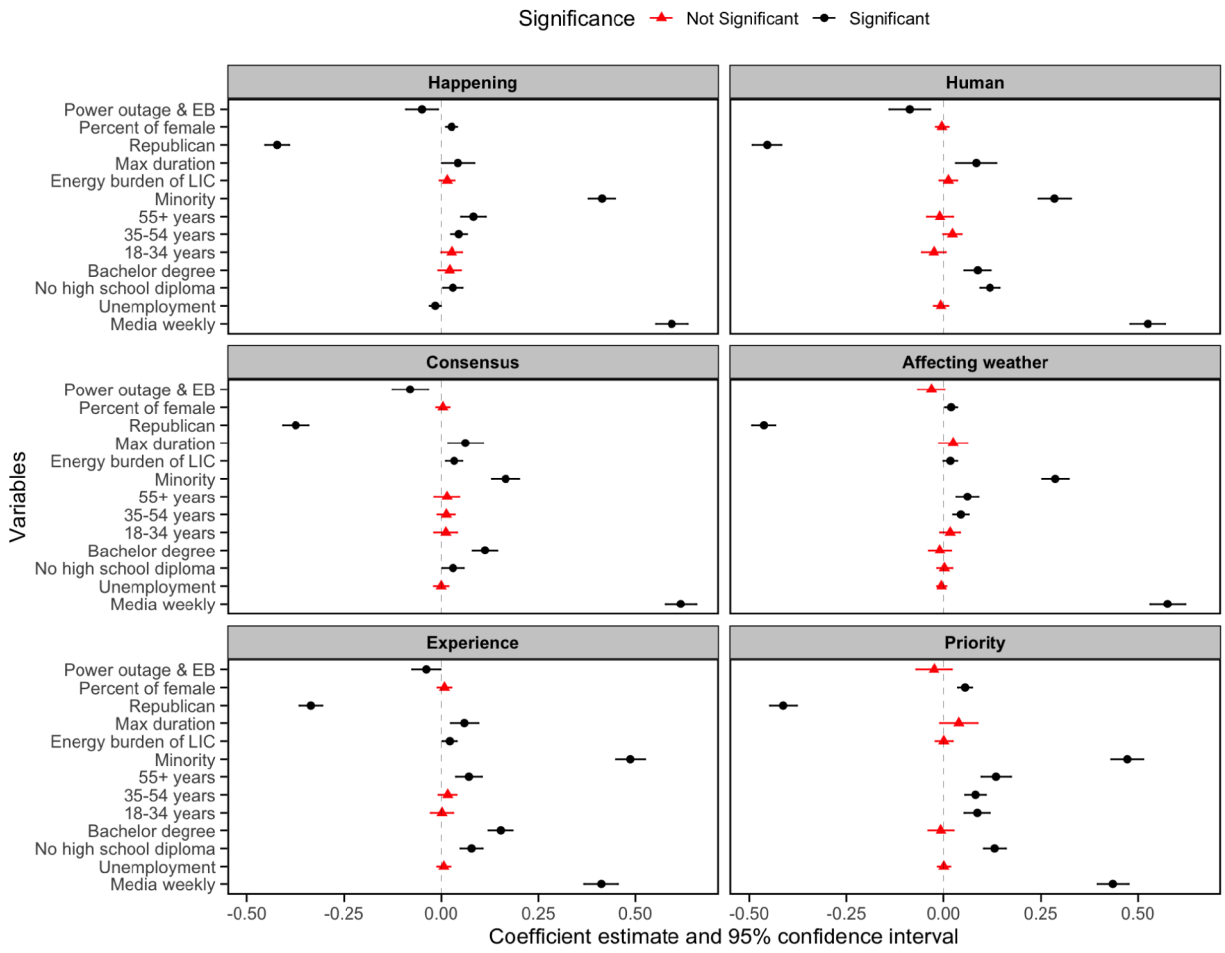


Figure 6. Associations between six climate change belief indicators and 13 independent variables. Points are adjusted coefficient estimates from our geospatial regression model. Error bars correspond to 95% CIs. Dotted lines denote the null of no association. Red bars indicate that the corresponding independent variable is not significantly correlated with the climate change belief indicator, while black bars indicate a significant correlation. EB represents energy burden. Max duration is maximum power outage durations. Power outage & EBs is the interaction term between power outages and EBs of low-income communities. The dependent variables corresponding to the six subfigures are as follows: (1) Happening; (2) Human; (3) Consensus; (4) Affecting weather; (5) Experience; (6) Priority.

consumption, and minority populations suggest the roles of power system infrastructure resilience, targeted communication strategies, and community engagement as potential avenues for enhancing climate change awareness. Conversely, the strong negative association with political affiliation highlights the challenge of polarization. These insights contribute to a deeper understanding of the geospatial dynamics of climate change beliefs and emphasize the need to account for the local context and spatial autocorrelation when examining public perceptions. Policy efforts that strengthen community resilience, alleviate EBs, and enhance grid reliability may be critical for advancing the public understanding of climate issues. Reaching diverse communities requires recognizing the differential exposure to systemic inequities and climate impacts.

Nevertheless, this study has limitations. Although our county-level geospatial regression analyses yield relatively high R^2 values (~0.9), the cross-sectional design limits causal inference. Findings should be interpreted as associations between county-level contextual conditions (e.g., outage exposure, EBs), demographic composition, and aggregate belief patterns—not as causal effects on individuals or places. Potential endogenous relationships may not have been fully accounted for, and unobserved confounding variables could influence the observed associations. Future research should address this by incorporating more longitudinal data or applying spatially explicit causal inference frameworks. Last but not least, even though the climate change beliefs from YCOM are multidimensional in scope, widely used, and capture broad geographic variation, they have important limitations. As a survey-based product, they are vulnerable to sampling, nonresponse, and measurement biases that can constrain their generalizability. The smaller-scale estimates are produced by statistical downscaling: even with large individual-level samples and geographic, demographic, and economic covariates, uncertainty increases at finer spatial scales and remains nontrivial for counties. Results could also shift with alternative geographic aggregations (the modifiable areal unit problem, MAUP), and county averages are ecological or area-level summaries that do not represent individual attitudes. Because these are modeled estimates rather than direct observations, they may not fully capture local infrastructure stressors (e.g., grid reliability and housing quality) that may shape beliefs. To limit overinterpretation, we do not rank counties by point estimates; instead, we group results into

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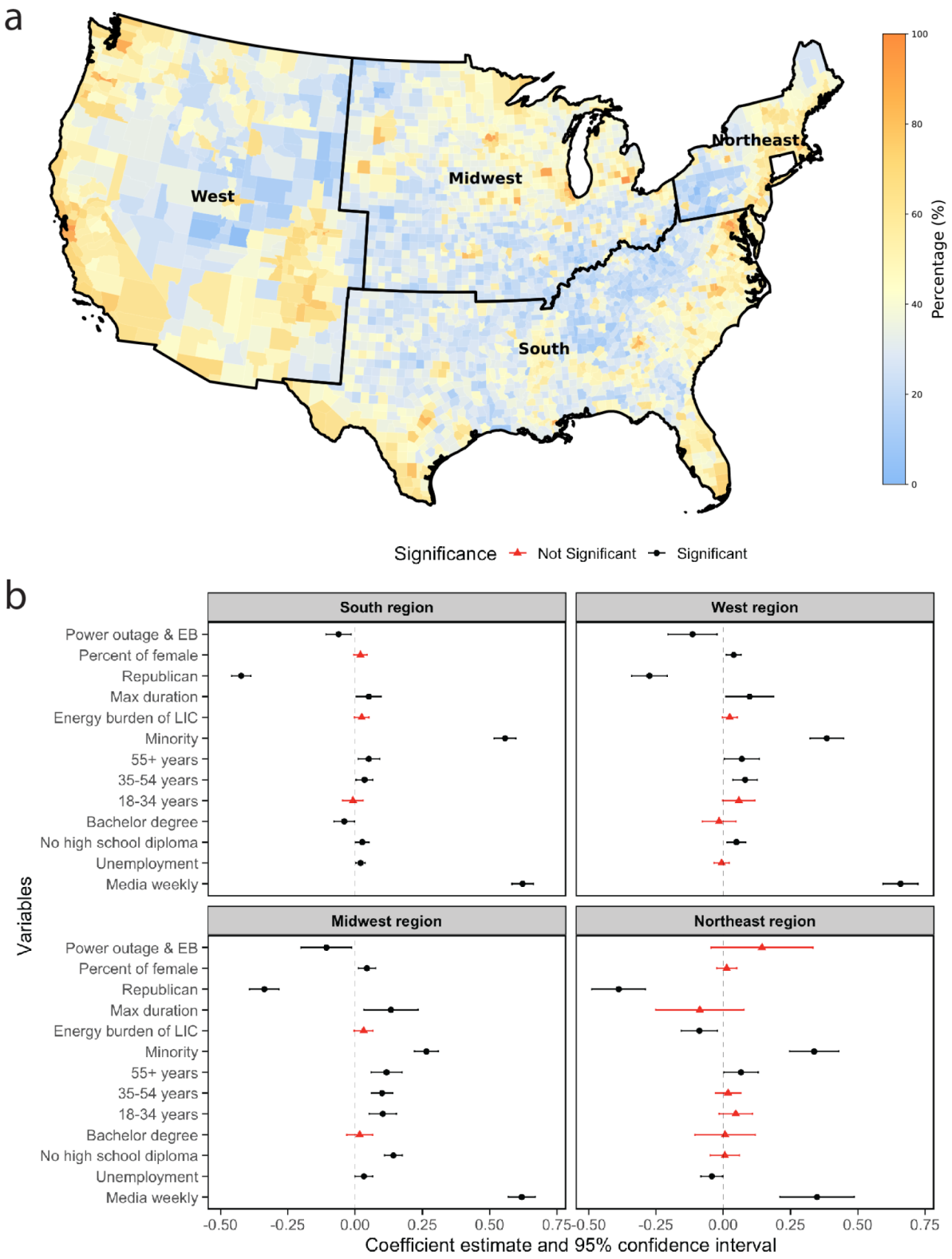


Figure 7. Regional distributions of climate change beliefs and their association with independent variables. (**a**) County-level climate change belief metric distribution in the continental U.S. The metric is aggregated from the six indicators (see eq 1). (**b**) Association between climate change beliefs and the independent variables for the South, West, Midwest, and Northeast regions. Points are adjusted coefficient estimates from geospatial regression models. Error bars correspond to 95% CIs. Dotted lines denote the null of no association. Red bars indicate that the corresponding independent variable is not significantly correlated with the climate change belief metric, while black bars indicate a significant correlation. EB is energy burden. Max duration is maximum power outage durations. Power outage & EB is the interaction term between power outages and EB of low-income communities.

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Environ. Sci. Technol. XXXX, XXX, XXX−XXX

quintiles and treat small differences as indistinguishable given measurement error and model uncertainty.

Moreover, our study relied on stated-preference techniques and self-reported data on outages and beliefs. Future research could explore beliefs and attitudes with more integrative and coproduced forms of data collection and research. Smith et al.[77] suggest that one option could be participatory action research or community-engaged research, where members of the public or vulnerable groups are actively involved in the research process. Future research could utilize citizen science techniques or community science based on local knowledge, rather than academic or expert knowledge. It could also harness ethnographic or photovoice approaches that empower participants to document their lived experiences through visual media.

Finally, a critical consideration in the future is the potential endogeneity of power outage frequency. Regions experiencing frequent outages may possess unobserved characteristics, such as infrastructural vulnerabilities or socioeconomic factors, that simultaneously influence both the frequency of blackouts and residents' climate change beliefs. If unaddressed, this endogeneity could bias regression estimates. To mitigate this concern, incorporating exposure to extreme weather events as a control variable of geospatial regression models is advisable. Including such variables can help determine whether weather events are potential confounders influencing both blackout occurrences and climate change beliefs, thereby isolating the effect of blackouts while holding other factors constant. Alternatively, extreme weather exposure could serve as an independent variable (IV), provided it is correlated with power outages but uncorrelated with the geospatial regression model's error term. Another approach is to include an interaction term between power outage duration and weather exposure to test whether the effect of outages on climate change beliefs varies by exposure level. This would help assess whether individuals in frequently affected regions respond differently to power disruptions when forming climate change beliefs. Moreover, while our study highlights the role of EBs in low-income communities, further investigations should consider broader conceptualizations of energy insecurity, precariousness, and vulnerability, which can include housing quality, access to energy-efficient technologies, and social support systems, or even other nondemographic factors such as spatial injustice, temporal injustice, or impacts on nonhuman forms of life.[78] Understanding these multifaceted factors will provide a more comprehensive view of how energy experiences shape climate perceptions and vice versa.

6. IMPLICATIONS

This study provides robust evidence of a significant association among power outages, EBs, and climate change beliefs across the U.S. Temporally, we observed that changes in beliefs are not linearly associated with changes in power outages. Our geospatial analysis finds that counties with longer maximum outage duration are consistently linked to stronger climate change beliefs, with each one-unit increase in maximum outage duration associated with a 5.16% rise in beliefs. These findings highlight how direct experiences with infrastructure disruption can shape public perceptions of climate risk. Regional heterogeneity in these patterns highlights the importance of the local context. Crucially, the interaction term between power outages and EBs of LICs showed a significant negative association with beliefs. This negative association reveals that counties with higher energy costs and prolonged outages may perceive climate challenges differently, highlighting the importance of considering both economic precarity and lived infrastructure experiences when evaluating public engagement with climate issues. Moreover, the observed associations with political affiliation, education, and minority status are consistent with prior research, reinforcing the complex and multifactorial nature of climate change beliefs.

This research offers novel insights into how climate change beliefs at a county level are influenced not only by socioeconomic, demographic, and ideological factors but also by lived experience, particularly through the lens of energy insecurity. The findings underscore the need for integrated, equity-focused policies that improve grid reliability, reduce EBs, and strengthen community resilience. Doing so can not only enhance energy justice but also deepen public engagement with climate change in an era of growing energy deprivation, coupled with climatic and environmental disruption.

■ ASSOCIATED CONTENT

Data Availability Statement

All data used in the analysis and to create Figures 2−7 will be available publicly from the following sources: • Climate Change Beliefs Data from YCOM: https://climatecommunication.yale.edu/visualizations-data/ycom-us-2024/ • CDC/ATSDR Social Vulnerability Index (SVI) Database: https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html • EAGLE-I Power Outage Database Developed by Oak Ridge National Lab: https://www.osti.gov/dataexplorer/biblio/dataset/1975203 • Energy Burden Data from LEAD Tool: https://catalog.data.gov/dataset/low-income-energy-affordability-data-lead-tool-2022-update • Republican Candidate-Votes Database: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VOQCHQ • American Community Survey (ACS) Bachelor Percentage Data: https://data.census.gov/table/ACSST5Y2022.S1501?g=860XX00US14067 • American Community Survey (ACS) Age, Gender, and Race Data: https://data.census.gov/table/ACSST5Y2023.S0101?g=860XX00US06614

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.5c09116.

Supplementary figures (climate change belief stratification, outage event definition and processing, spatial diagnostics including Moran's I, hot-spot maps, correlation matrix, VIF plots); supplementary tables (state-level outage statistics, annual outage characteristics, full regression coefficients and diagnostics); extended methodological justification for spatial regression model selection (PDF)

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https://doi.org/10.1021/acs.est.5c09116
Environ. Sci. Technol. XXXX, XXX, XXX−XXX

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Author Contributions

H.S. designed the research framework, processed county-level data, conducted regression analyses and data visualization, and led the writing of the manuscript. C.-F.C. conceptualized the study, secured funding, and contributed to the analysis and interpretation of results and writing. B.S. contributed to the analysis and interpretation of results and assisted with writing the Introduction, Discussion, and Methods sections. S.S. assisted with the literature review and contributed to the writing of the Introduction, Results, and Discussion sections. Z.S. curated and processed county-level data on power outages and assisted with data visualization. All authors discussed the results and wrote and reviewed, revised, and edited the manuscript.

Notes

Code Availability: The code utilized in the study is available from the corresponding author upon reasonable request.

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Chien-Fei Chen and Hang Shuai acknowledge the support from Wellcome Trust Foundation grant number #227151/Z/23/Z, United Kingdom.

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